

## Day 9 :

# Task 1: Monitor the resources during ARM template deployment.

The screenshot shows the Microsoft Azure portal interface. On the left, the 'Resource Manager | Resource groups' sidebar is visible, with 'FardeenAttar-rg' selected. The main pane displays the 'FardeenAttar-rg | Activity log' page. The 'Overview' tab is active, showing a list of 69 items. The table columns are: Operation name, Status, Time, Time stamp, Subscription, and Event initiated by. The table lists various operations such as 'Health Event Updated', 'Delete Storage Account', 'Get deployment operation status', and 'Export template for deployment', all with a status of 'Succeeded'.

The screenshot shows the 'FardeenAttar-rg | Deployments' page in the Microsoft Azure portal. The 'Deployments' tab is active, displaying a table of deployment operations. The table columns are: Deployment name, Status, Last modified, Duration, and Related events. The table lists various deployments such as 'PolicyDeployment\_1238214245253546518', 'azuredploy', 'storage', 'acr', 'instance', 'disk', 'fardeen313\_1770793608978', 'ContainerInstanceDeployment-20260210162235', 'NoMarketplace-20260210161955', 'Microsoft.ContainerRegistry', 'network-interface-associated-virtual-network-202...', 'CreateVm-canonical.ubuntu-24\_04-lts-server-2026...', 'Microsoft.CustomScriptExtension1', 'network-interface-associated-virtual-network-202...', 'CreateVm-microsoftwindowsdesktop.windows-11...', 'fardeen313\_1770634878157', 'network-interface-associated-virtual-network-202...', and 'network-interface-associated-virtual-network-202...'. All deployments have a status of 'Succeeded'.

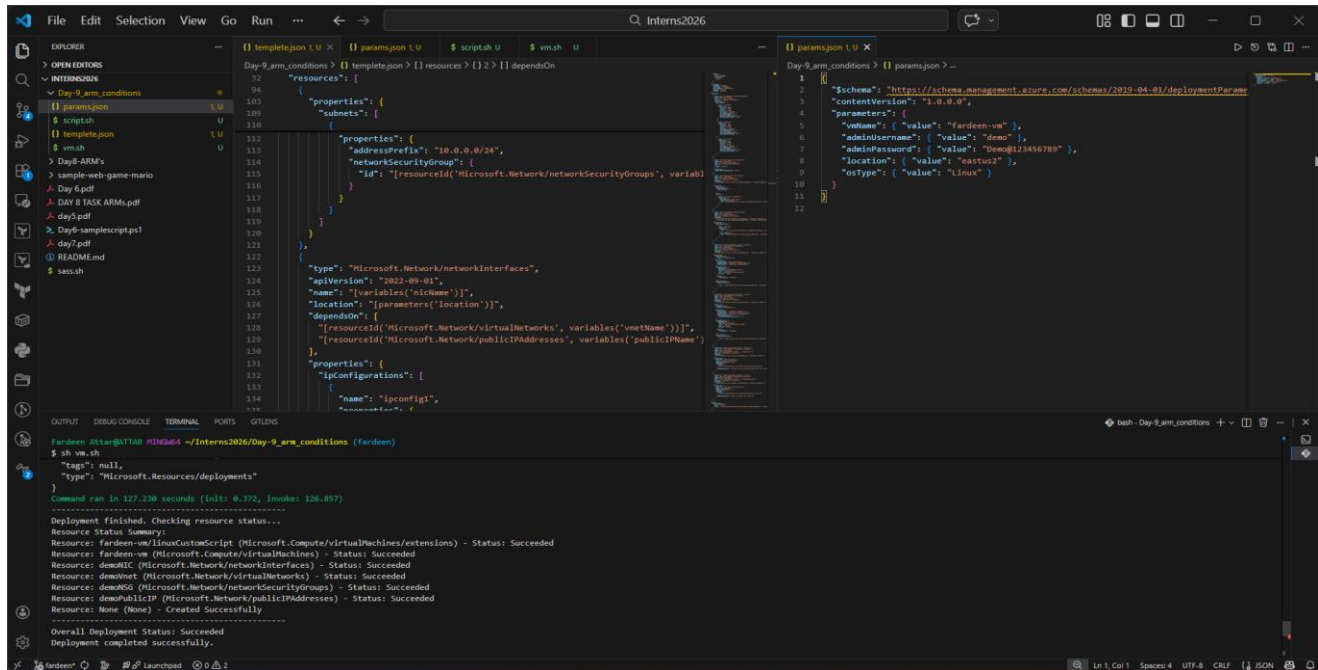
```
Deployment Finished. Checking Status...
Result
-----
Succeeded
-----

Succeeded
Listing Resource Operations...
-----
Succeeded
Listing Resource Operations...
-----
Succeeded
Listing Resource Operations...
OperationId      ResourceGroup
-----
8D7B9F0C1A574719 FardeenAttar-rg
75409B5FFE4EF1B8 FardeenAttar-rg
08584307188397470759 FardeenAttar-rg

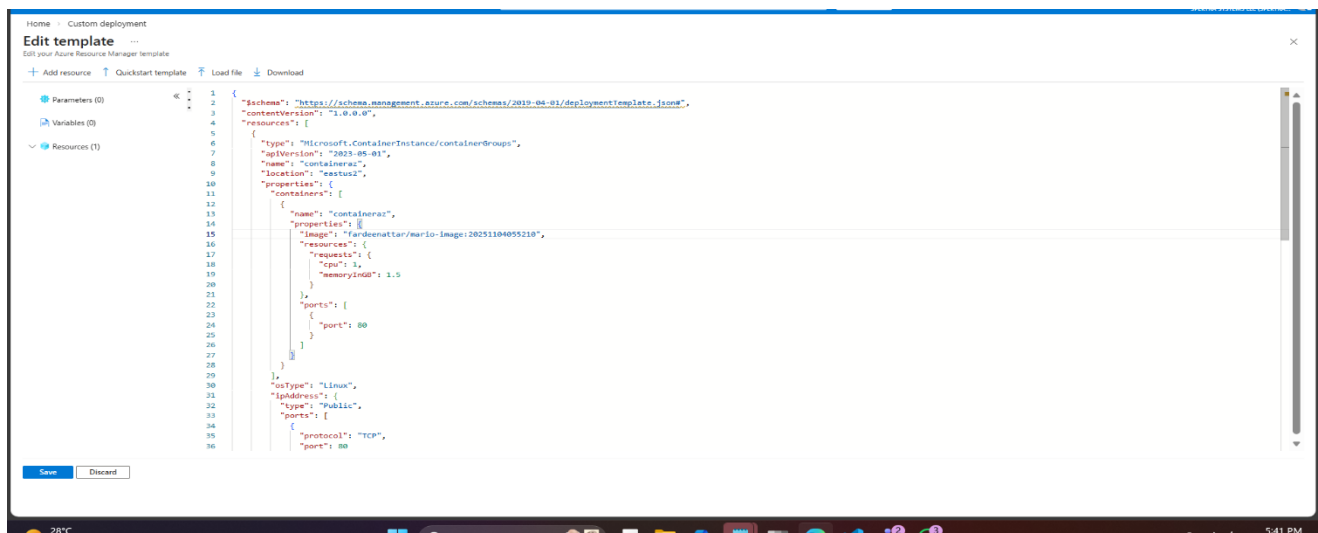
Fardeen Attar@ATTAR MINGW64 ~/Interns2026/Day8-ARM's/StorageAccount (fardeen)
$
```

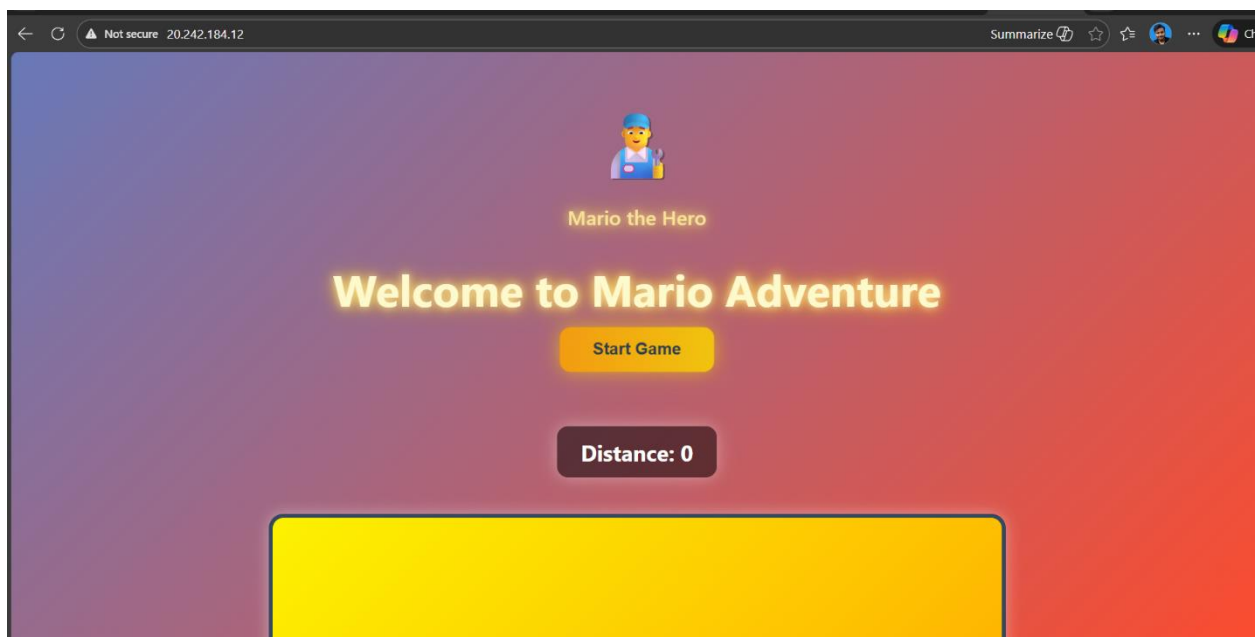
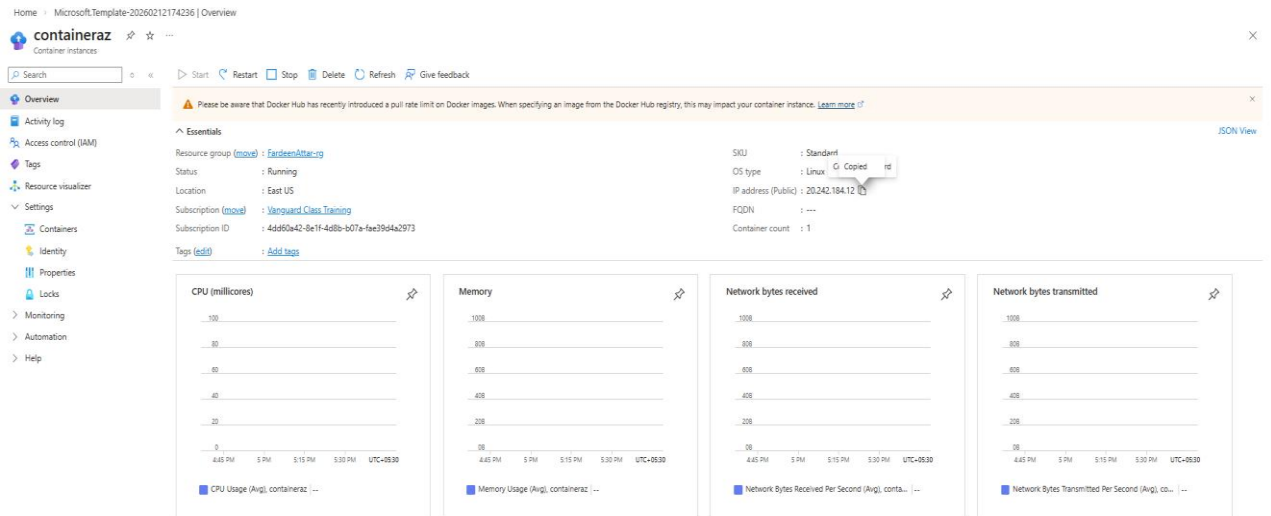
Task 2: Explore the different ARM deployment methods and the ARM template writing process.

- Explored different ARM deployment methods in Azure.
- Deployed an Azure resource ARM,s using Azure CLI (Git Bash).



- Deployed the Azure resource using Azure Portal Custom Deployment with a custom ARM template.





- Reviewed ARM documentation to understand template structure and deployment process.

Task 3: Created Windows and Linux VMs using a single ARM template with condition blocks, configured Custom Script Extensions, and deployed IIS on Windows and Nginx on Linux.

## 1-Linux

The image shows the deployment of a Linux VM using an ARM template. The top section displays the Visual Studio Code interface with the ARM template editor. The template defines resources for a Linux VM, network interfaces, and public IP addresses. The deployment is successful, as indicated by the output window.

```
template.json
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentParameters.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "vmName": { "value": "fardeen-vm" },
    "adminUsername": { "value": "demo" },
    "adminPassword": { "value": "Demo@123456789" },
    "location": { "value": "eastus2" },
    "osType": { "value": "linux" }
  }
}
```

The output window shows the deployment details:

```
Fardeen Attar@ATTAR: ~/Interns2026/Day-9_arm_conditions (fardeen)
$ sh vm.sh
"tags": null,
"type": "Microsoft.Resources/deployments"
}
Command ran in 127.230 seconds (init: 0.372, invoke: 126.857)
Deployment finished. Checking resource status...
Resource Status Summary:
Resource: fardeen-vm/LinuxCustomScript (Microsoft.Compute/virtualMachines/extensions) - Status: Succeeded
Resource: fardeen-vm (Microsoft.Compute/virtualMachines) - Status: Succeeded
Resource: demoNIC (Microsoft.Network/networkInterfaces) - Status: Succeeded
Resource: demoVnet (Microsoft.Network/virtualNetworks) - Status: Succeeded
Resource: demoNSG (Microsoft.Network/networkSecurityGroups) - Status: Succeeded
Resource: demoPublicIP (Microsoft.Network/publicIPAddresses) - Status: Succeeded
Resource: None (None) - Created Successfully
Overall Deployment Status: Succeeded
Deployment completed successfully.
```

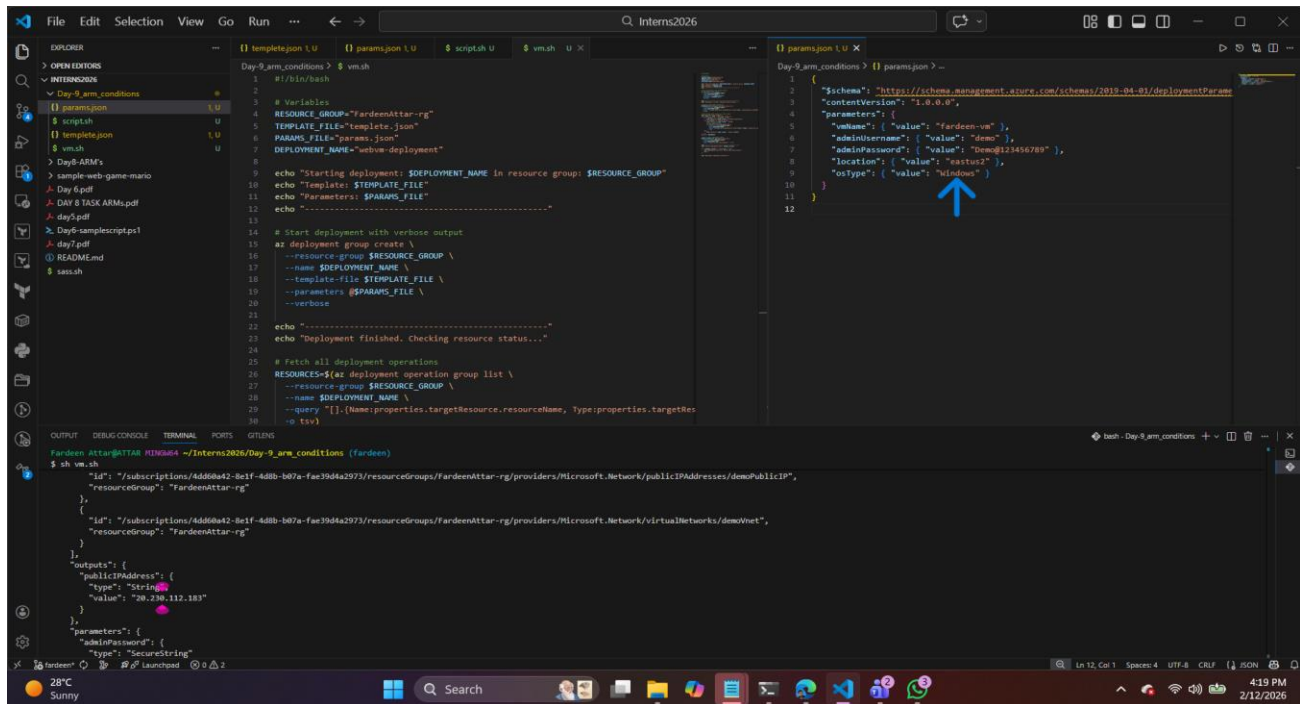
The bottom section shows a web browser displaying the deployed web application. The page has a blue background and contains the following text:

# Welcome to Fardeen's WebApp!

This page is deployed via ARM template and Nginx.

Enjoy your automated deployment!

## 2-Windows



```
1 #!/bin/bash
2
3 # Variables
4 RESOURCE_GROUP="FardeenAttar-rg"
5 TEMPLATE_FILE="template.json"
6 PARAMS_FILE="params.json"
7 DEPLOYMENT_NAME="winvm-deployment"
8
9 echo "Starting deployment: $DEPLOYMENT_NAME in resource group: $RESOURCE_GROUP"
10 echo "Template: $TEMPLATE_FILE"
11 echo "Parameters: $PARAMS_FILE"
12 echo "-----"
13
14 # Start deployment with verbose output
15 az deployment group create \
16 --resource-group $RESOURCE_GROUP \
17 --name $DEPLOYMENT_NAME \
18 --template-file $TEMPLATE_FILE \
19 --parameters @$PARAMS_FILE \
20 --verbose
21
22 echo "-----"
23 echo "Deployment finished. Checking resource status..."
24
25 # Fetch all deployment operations
26 RESOURCES=$(az deployment operation group list \
27 --resource-group $RESOURCE_GROUP \
28 --name $DEPLOYMENT_NAME \
29 --query "[].(Name:properties.targetResource.resourceName, Type:properties.targetRes
30 -p tsv)
```

```
Fardeen Attar@M17AR RIN364 ~/Interns2026/Day-9_arm_conditions (fardeen)
$ sh vm.sh
{"id": "/subscriptions/4dd0a42-be1f-4d8b-b07a-fae39da2973/resourcegroups/fardeenattar-rg/providers/Microsoft.Network/publicIPAddresses/demPublicIP",
  "resourceGroup": "FardeenAttar-rg"
},
{
  "id": "/subscriptions/4dd0a42-be1f-4d8b-b07a-fae39da2973/resourcegroups/fardeenattar-rg/providers/Microsoft.Network/virtualNetworks/demVnet",
  "resourceGroup": "FardeenAttar-rg"
},
  "outputs": {
    "publicAddress": {
      "type": "String",
      "value": "20.236.112.183"
    }
  },
  "parameters": {
    "adminPassword": {
      "type": "SecureString"
    }
  }
}
```

